

AKSHAY MAHALE

AI/ML Engineer

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PROFILE SUMMARY

AI/ML Data Scientist with 4+ years of experience in Machine Learning, Deep Learning, NLP, and Generative AI. Proficient in building end-to-end ML/DL pipelines, developing chatbots using Retrieval-Augmented Generation (RAG), and implementing time-series forecasting solutions. Strong expertise in Python, Scikit-learn, TensorFlow, PyTorch, Azure AI/ Search, Hugging Face, Graph API, n8n and LangChain. Proven ability to deliver AI solutions that improve operational efficiency, customer satisfaction, and decision-making.

WORK EXPERIENCE

MOBICOOL SOFTWARE SOLUTIONS PVT LTD - DATA SCIENTIST

March 2022 - Present

Developed and implemented AI solutions leveraging the RAG (Retrieval-Augmented Generation) framework, transforming large-scale unstructured enterprise data into actionable insights for internal and client-facing applications.

Built and optimized NLP pipelines involving text cleaning, tokenization, embedding generation, and semantic retrieval to support intelligent knowledge systems and document understanding workflows.

Managed end-to-end data pipelines, including data ingestion, preprocessing, workflow automation, vector storage, and semantic search integration using tools such as Microsoft Graph API, n8n, and Qdrant, ensuring accuracy, scalability, and data integrity.

Collaborated with cross-functional teams to enhance AI-driven automation, improve information accessibility, streamline workflows, and support data-driven decision-making across multiple departments.

TECHNICAL SKILLS

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| <ul style="list-style-type: none">• Python• Prediction / Forecasting• Numpy• MySQL• Data Analyst• Ai Agents• Pytorch• Transformers• PEFT, LoRA, QLoRA• AWS EC2 Sagemaker S3• MCP• n8n | <ul style="list-style-type: none">• Machine Learning• NLP / NLU• OOPs• MSSQL• RAG• Langchain• Fast API• Vector Databases• Prompt Engineering• LLaMA4• Fine-Tuning• MS Graph API | <ul style="list-style-type: none">• Deep Learning• Pandas• Chat bot• LLM• Time Series Analysis• Langgraph• Scikit Learn• GPT• Git GitHub• Hugging Face Pipeline• Transformer• Azure AD |
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PROJECTS

AI-CONTEXTUAL MEDICAL QUERY FRAMEWORK

DOMAIN - HEALTH CARE

Implemented an end-to-end RAG (Retrieval-Augmented Generation) framework to build an intelligent query response system, combining retrieval and generation capabilities.

Developed and optimized pipelines for data ingestion, chunking, semantic embeddings (using Sentence-BERT), and vector storage in Pinecone to enable efficient and accurate semantic search. Integrated Llama 2 for contextual answer generation, fine-tuned for domain-specific responses, and enhanced system performance using Evaluation metrics like BLEU, ROUGE, precision, and recall.

Skills: Python | Chatbot | NLP | Machine Learning | Deep Learning | LLM | RAG | AWS

AI-FINANCIAL AID KNOWLEDGE ASSISTANT SYSTEM

DOMAIN - EDUCATION

Developed an end-to-end RAG (Retrieval-Augmented Generation) pipeline to transform enterprise SharePoint content into an intelligent, query-responsive knowledge assistant for internal staff.

Built automated pipelines for document crawling, cleaning (ASPX- Markdown), chunking, semantic embeddings, and vector storage in Qdrant to enable accurate semantic search across large, heterogeneous document sets.

Integrated workflows using n8n, Microsoft Graph API, and Azure AD authentication to enable scheduled ingestion, monitoring, and automated updates, improving assistant reliability and knowledge freshness.

Skills: Python | Generative AI | Qdrant | n8n | MS Graph API | LLM

CUSTOMER RETENTION ANALYSIS

DOMAIN - TELECOMMUNICATION

Developed predictive models with 85% accuracy to identify high-risk customers, achieving a 20% reduction in churn over 12 months. Implemented cost-effective retention strategies, lowering the annual churn rate by 15% and enhancing customer loyalty. Optimized customer acquisition costs by prioritizing satisfaction and long-term retention.

Managed data collection, preprocessing, feature engineering, and imbalanced dataset handling (using SMOTE) for robust predictive analysis, enabling actionable insights from structured data.

Designed and fine-tuned Random Forest models using hyperparameter tuning and regularization techniques to achieve optimal performance. Deployed the solution on AWS EC2, ensuring scalability and real-time predictions.

Skills: Python | Predictive Modelling | Machine Learning | Pandas | Data Analysis.

EDUCATION

BACHELOR OF ENGINEERING - CGPA : 7.46

June 2021

Savitribai Phule Pune University

CERTIFICATIONS

- Supply chain planning and scheduling at TCS, Coursera
- Applied generative AI , Simplilearn.